Instructions:

- **Due:** February 9, 11:59PM . Upload one **single PDF** to Blackboard containing your complete submission and upload all associated LaTeX source files.
- Each problem is worth $\dfrac{100}{\text{number of problems}}$ points.
- Start each problem on a new page.
- Label your PDF file exactly as

  `ENME408_S26_HW02_YourLastName.pdf`

- All work must be typeset in LaTeX. Handwritten or scanned submissions will not be accepted.
- Failure to follow these submission and formatting guidelines may result in a substantial point deduction or a score of zero.

Problem 1. (Unit cube vectors.) Consider a unit cube with one corner at $(0, 0, 0)$ and edges along the coordinate axes.

1. List all 8 corners of the cube as vectors in $\mathbb{R}^3$.
2. Let a be the vector from $(0, 0, 0)$ to $(1, 0, 0)$ and let b be the vector from $(0, 0, 0)$ to $(0, 1, 0)$. Compute $a \times b$ and interpret the result.

Problem 2. Use the dot product to compute the angle between the vectors $x = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and $y = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$.

Problem 3. Are the vectors in Problem 2 orthogonal? Justify your answer.

Problem 4. Let $x = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$ and $y = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

1. Compute $x \bullet y$.
2. Compute $\|x\|$ and $\|y\|$.
3. Compute the angle θ_{xy} using $\theta_{xy} = \cos^{-1}\left(\dfrac{x \bullet y}{\|x\|\|y\|}\right)$.

Problem 5. Let $x = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. Find *two distinct* nonzero vectors $u_1, u_2 \in \mathbb{R}^2$ such that $u_1 \bullet x = 0$ and $u_2 \bullet x = 0$. Then compute $\theta_{u_1 x}$ and $\theta_{u_2 x}$ to verify orthogonality.

Problem 6. Let $x, y \in \mathbb{R}^n$ with $x \neq 0$. Define the projection of y onto x by

$$\text{proj}_x(y) \triangleq \left(\frac{x \bullet y}{x \bullet x}\right) x.$$

1. Show that $y - \text{proj}_x(y)$ is orthogonal to x.
2. Interpret this result geometrically.

Problem 7. Let $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$. Show that $(x \times y) \bullet x = 0$ and $(x \times y) \bullet y = 0$.

Problem 8. Let $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$. Lagrange's identity states that

$$\|x \times y\|^2 = \|x\|^2 \|y\|^2 - (x \bullet y)^2.$$

Use this identity to show that

$$\|x \times y\| = \|x\| \|y\| |\sin \theta_{xy}|.$$

Problem 9. Let $x = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix}$ and $y = \begin{bmatrix} 1 \\ 5 \\ -2 \end{bmatrix}$.

1. Compute $x \times y$.
2. Compute $\|x \times y\|$, $\|x\|$, $\|y\|$, and $x \bullet y$.
3. Verify Lagrange's identity

$$\|x \times y\|^2 = \|x\|^2 \|y\|^2 - (x \bullet y)^2.$$

Problem 10. Let $x, y \in \mathbb{R}^n$. Prove the Cauchy–Schwarz inequality

$$|x \bullet y| \leq \|x\| \, \|y\|.$$

Problem 11. Let

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}.$$

Compute the matrix products AB and BA. Are these two products equal? Briefly explain why or why not.

Problem 12. Consider the matrices

$$A = \begin{bmatrix} 3 & 0 & 2 \\ 4 & 3 & 5 \\ 7 & 2 & 6 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 5 \\ 6 & 7 \end{bmatrix}.$$

1. Manually compute $\det(A)$ using a cofactor expansion.
2. Manually compute B^{-1} using the standard formula for the inverse of a 2×2 matrix.

You may use MATLAB to check your numerical results, but your solution must clearly show the hand calculations.

Problem 13. Consider the linear system:

$$2x_1 + 3x_2 - x_3 = 4, \tag{1}$$
$$-x_1 + 4x_2 + 2x_3 = -3, \tag{2}$$
$$3x_1 - 2x_2 + 4x_3 = 5. \tag{3}$$

1. Write this system in matrix form $Ax = y$, where

$$x \triangleq \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix},$$

   and explicitly identify the matrix $A \in \mathbb{R}^{3\times 3}$ and the vector $y \in \mathbb{R}^3$.
2. Verify using MATLAB that $\det(A) \neq 0$.
3. Compute A^{-1} (by hand or using MATLAB) and solve for x using

$$x = A^{-1} y.$$

Problem 14. Consider the matrices

$$A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}, \qquad B = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}. \tag{4}$$

1. Show analytically that A is an orthogonal matrix by verifying that

$$A^\top A = I.$$

2. Determine whether B is orthogonal using the same criterion, and justify your answer.

3. Let $x \triangleq \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. For each angle

$$\theta \in \left\{ \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4} \right\},$$

compute $y = Ax$ and determine the angle θ_{xy} between x and y (express your answer in terms of π).

Problem 15. Let A be the matrix from Problem 12.

1. Use MATLAB's `eig` command to compute the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of A.

2. Using these eigenvalues, verify manually that

$$\det(A) = \lambda_1 \lambda_2 \lambda_3, \tag{5}$$
$$\operatorname{tr}(A) = \lambda_1 + \lambda_2 + \lambda_3. \tag{6}$$

You may use your determinant from Problem 12.

3. Use MATLAB to compute:
   - the eigenvalues $\lambda_{\text{sq},i}$ of A^2, and
   - the eigenvalues $\lambda_{\text{inv},i}$ of A^{-1}.

4. Explain how $\lambda_{\text{sq},i}$ and $\lambda_{\text{inv},i}$ relate to $\lambda_1, \lambda_2, \lambda_3$.